

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
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1 Objective

The goal of this experiment is to group Human Activity Recognition data into clusters using three methods. The methods are K-Means, DBSCAN, and Hierarchical Clustering. We compare how well each method separates the six activities without using the activity labels during training.

2 Problem Statement

Input: 561 numeric features per sample, describing accelerometer and gyroscope signals from a smartphone.

Output: A cluster label for each sample, found without using the true activity labels.

Objective: Find natural groups in the data and check how close these groups are to the real activities.

3 Dataset Description

Dataset Name	Human Activity Recognition Using Smartphones
Dataset Source	UCI Machine Learning Repository
Number of Samples	10299 (7352 train, 2947 test, combined for clustering)
Number of Features	561
Number of Classes	6 (Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, Laying)
Missing Values	0
Train-Test Split	70 percent train, 30 percent test in original data. Combined here since clustering does not need a split.

4 Exploratory Data Analysis

This section shows the shape of the data before clustering.

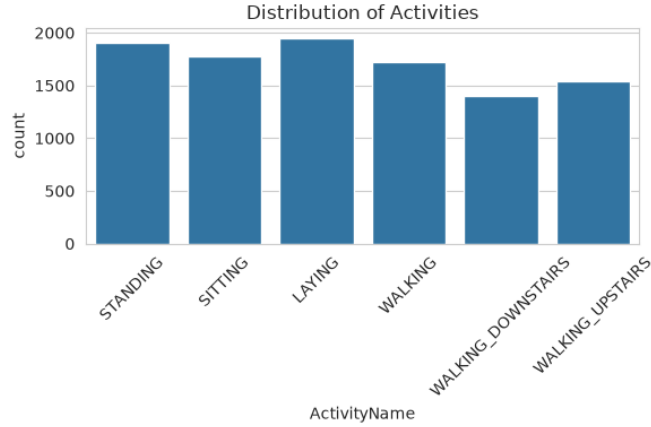


Figure 1: Distribution of activities in the dataset

Inference:

Each of the six activities has close to the same number of samples. There is no major class imbalance. This makes external metrics like ARI and NMI fair to compute.

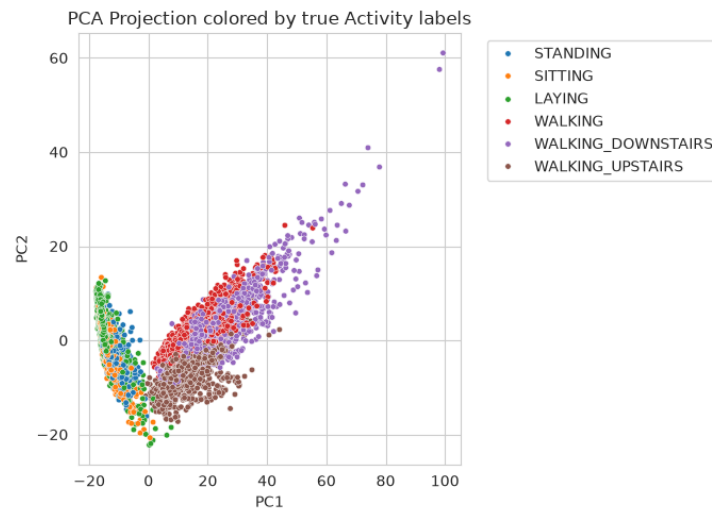


Figure 2: PCA projection colored by true activity labels

Inference:

Sitting, Standing, and Laying samples group together on the left. Walking, Walking Upstairs, and Walking Downstairs spread out on the right. This shows two broad natural groups: still activities and moving activities. Within the moving group, the three walking types overlap a lot.

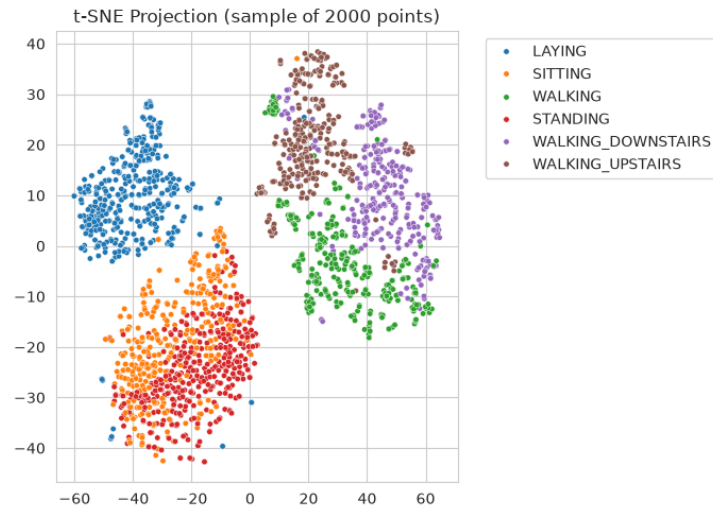


Figure 3: t-SNE projection on a sample of 2000 points

Inference:

t-SNE shows the same two broad groups seen in PCA. The still activities form a tight, separate patch. The moving activities form a larger, more spread out region. Sitting and Standing sit close together, which matches their similar posture.

5 Data Preprocessing

- Missing values: checked and none were found.
- Encoding: activity numbers were mapped to activity names for reference and evaluation.
- Feature scaling: all 561 features were standardized using StandardScaler, mean 0 and variance 1.
- Dimensionality reduction: PCA was applied to keep 95 percent variance, giving 104 components. This was used for clustering to reduce noise and speed up computation.
- Train-test split: not used for clustering. Train and test sets were combined into one dataset.

6 Machine Learning Algorithm

K-Means Clustering

Working principle: K-Means picks k centroids, assigns each point to the nearest centroid, then updates centroids as the mean of assigned points. This repeats until the centroids stop moving much.

Mathematical formulation: it minimizes the within cluster sum of squares, written as the sum over clusters of the squared distance between each point and its cluster centroid.

Advantages: simple, fast, and works well when clusters are round and similar in size.

Limitations: needs k to be chosen in advance, sensitive to outliers, and struggles with clusters of odd shapes.

DBSCAN

Working principle: DBSCAN groups points that are closely packed together. A point is a core point if it has at least `minPts` neighbors within distance `eps`. Clusters grow by connecting core points and their neighbors. Points that do not belong to any cluster are marked as noise.

Advantages: does not need `k` in advance, can find clusters of any shape, and can detect noise or outliers.

Limitations: sensitive to the `eps` and `minPts` values, and struggles when clusters have very different densities. This is a common issue in high dimensional data like this one.

Hierarchical Agglomerative Clustering

Working principle: each point starts as its own cluster. Pairs of clusters are merged step by step based on a linkage rule, until one cluster remains. Ward linkage merges the pair that increases the total within cluster variance the least.

Advantages: does not need `k` chosen before running, and the dendrogram shows the full merge structure.

Limitations: slower on large datasets, and once two points are merged the decision cannot be undone.

7 Experimental Setup

Python Version	3.12
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Linux
Hardware	Standard CPU, no GPU used

8 Hyperparameter Selection

Parameter	Value
K-Means: number of clusters (<code>k</code>)	2, chosen using the highest silhouette score
K-Means: <code>n_init</code>	10
DBSCAN: <code>eps</code>	15, chosen from the k-distance graph knee point
DBSCAN: <code>min_samples</code>	10
Hierarchical: linkage	Ward
Hierarchical: number of clusters	2, same as K-Means for fair comparison
PCA: variance retained	95 percent (104 components)

9 Results

Internal Metrics

Metric	Value
K-Means Silhouette	0.4125
K-Means Davies-Bouldin	1.0191
K-Means Calinski-Harabasz	8635.78
DBSCAN Silhouette	0.2522
DBSCAN Davies-Bouldin	0.9686
DBSCAN Calinski-Harabasz	75.39
Hierarchical Silhouette	0.4121
Hierarchical Davies-Bouldin	1.0200
Hierarchical Calinski-Harabasz	8611.76

External Metrics (compared to true activity labels)

Metric	Value
K-Means ARI	0.3296
K-Means NMI	0.5455
DBSCAN ARI	0.0009
DBSCAN NMI	0.0189
Hierarchical ARI	0.3325
Hierarchical NMI	0.5568

K-Means found 5 clusters plus noise points. Noise points were 1276 out of 10299 samples.

10 Result Visualization

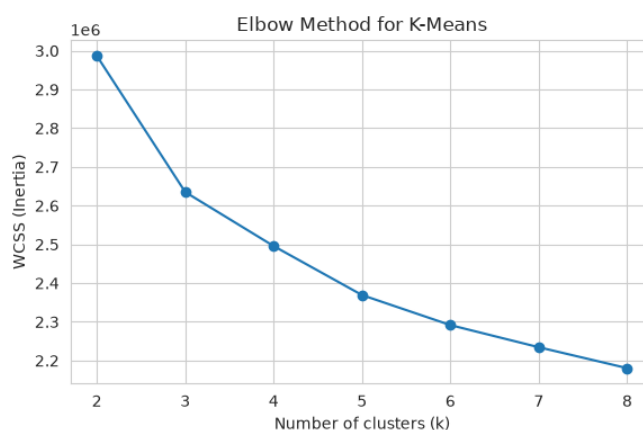


Figure 4: Elbow curve, k versus WCSS for K-Means

Inference:

WCSS drops fast from $k=2$ to $k=4$, then flattens slowly. There is no very sharp elbow point. This suggests the data does not split into many clean, equal sized clusters.

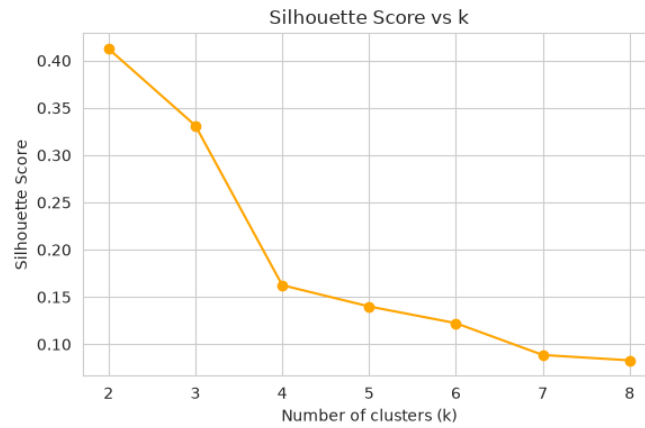


Figure 5: Silhouette score versus k for K-Means

Inference:

The silhouette score is highest at $k=2$ and drops as k increases. This means two clusters separate the data best by this internal measure. It matches the two broad groups seen earlier in PCA, still versus moving activities.

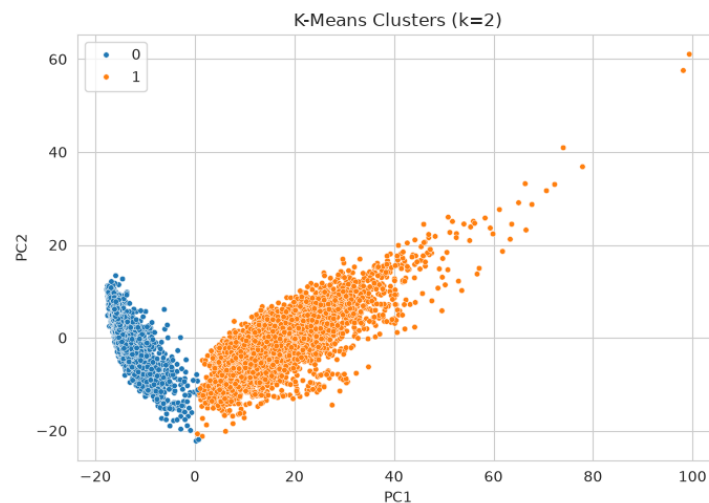


Figure 6: K-Means clusters with $k=2$ shown on PCA axes

Inference:

The two clusters closely match the still versus moving split. The boundary between clusters is fairly clean on the left side. Some overlap exists in the middle where walking types meet standing or sitting.

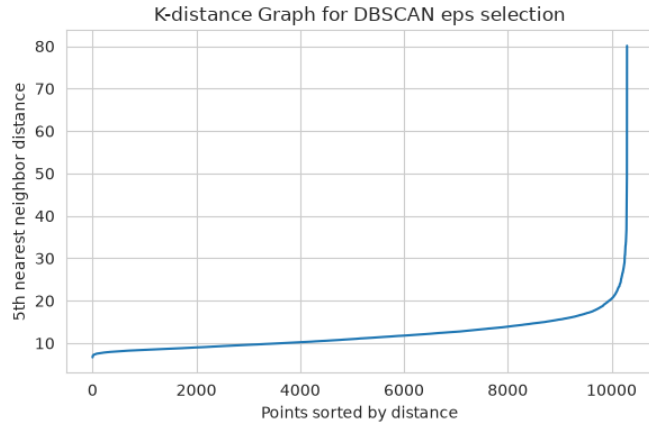


Figure 7: K-distance graph used to choose DBSCAN eps

Inference:

The curve rises slowly at first, then bends upward sharply near the end. The bend, or knee, marks a good eps value. An eps of 15 was chosen from this knee, close to where the curve starts rising fast.

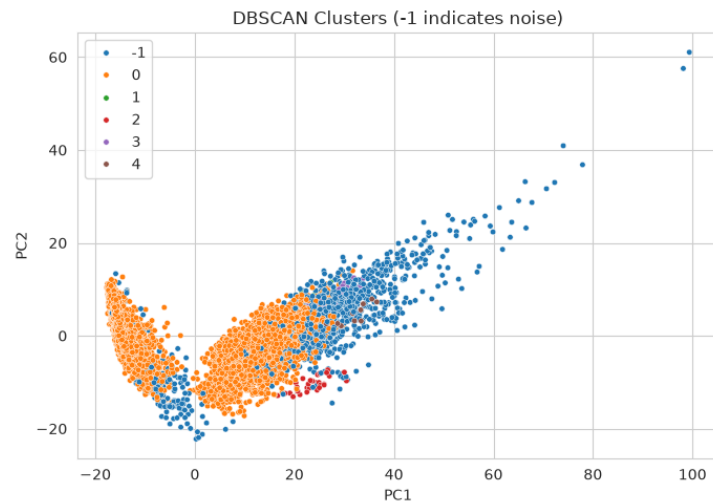


Figure 8: DBSCAN clusters, minus one means noise

Inference:

DBSCAN found 5 dense clusters and marked 1276 points as noise. One dense cluster covers almost all of the still activities on the left. The moving activities on the right are mostly marked as noise, with a few small dense pockets. This shows DBSCAN struggles when point density changes smoothly across a region.

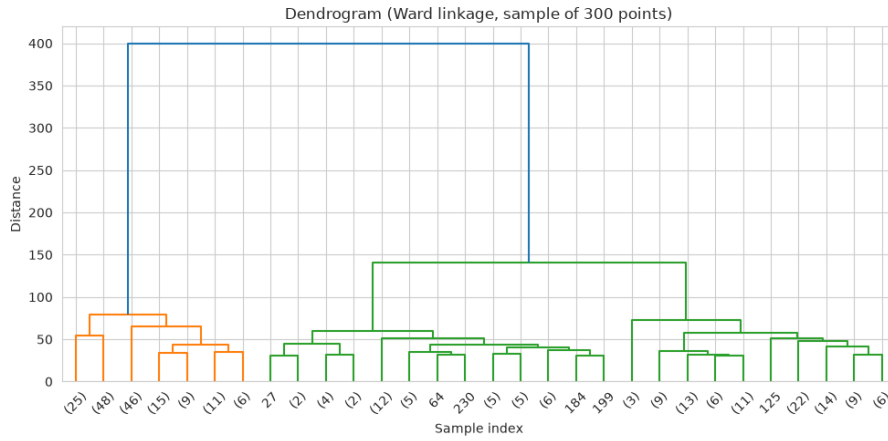


Figure 9: Dendrogram using Ward linkage, sample of 300 points

Inference:

The dendrogram shows two main branches splitting at the top. This matches the two cluster choice used for K-Means and Hierarchical. Smaller sub-branches lower down likely relate to the individual activities.

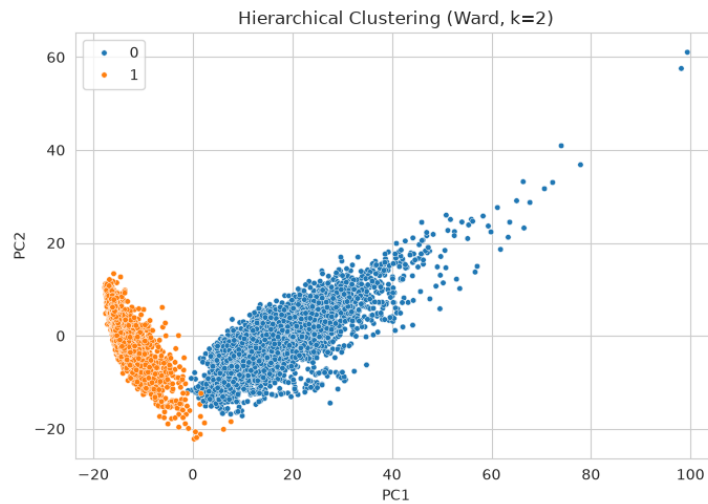


Figure 10: Hierarchical clustering with Ward linkage, k=2

Inference:

The result looks almost the same as the K-Means clusters. Both methods agree on the still versus moving split. This agreement is a good sign that the split is a real pattern in the data.

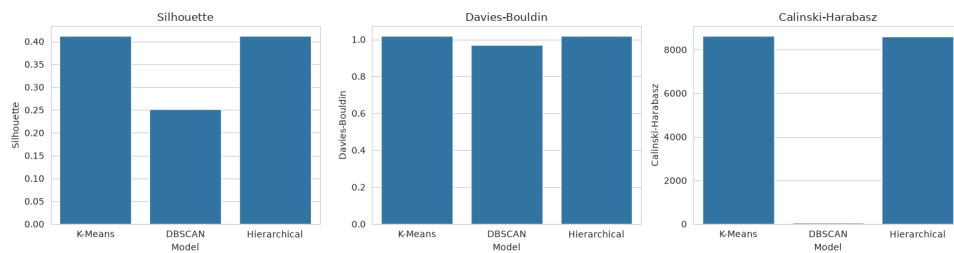


Figure 11: Bar plots of internal metrics across algorithms

Inference:

K-Means and Hierarchical have similar and higher silhouette scores. DBSCAN has a lower silhouette score but a slightly lower, better Davies-Bouldin score too. DBSCAN Calinski-Harabasz is much lower because it has more, smaller clusters plus noise.

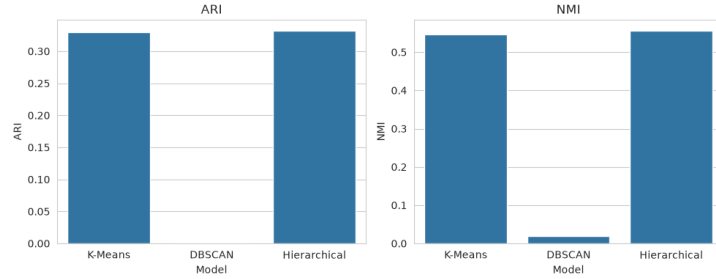


Figure 12: Bar plots of external metrics across algorithms

Inference:

K-Means and Hierarchical score similarly on ARI and NMI. DBSCAN scores close to zero on both external metrics. This means DBSCAN clusters do not line up well with the six true activity labels.

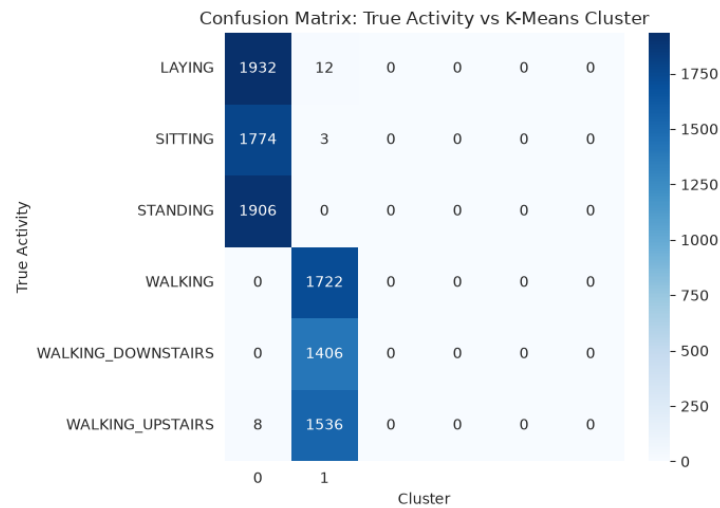


Figure 13: Confusion matrix, true activity versus K-Means cluster

Inference:

Cluster 0 mostly holds Sitting, Standing, and Laying samples. Cluster 1 mostly holds the three walking type activities. Very few samples from the still activities fall into the moving cluster. This confirms the two clusters mainly separate still activities from moving ones.

11 Discussion

K-Means and Hierarchical clustering gave almost the same result. Both split the data into still activities and moving activities. This split has a strong silhouette score, so it is a real, clear pattern in the data.

DBSCAN behaved differently. It found more, smaller clusters and marked some points as noise. Its silhouette score was lower, and it did not match the six true labels well. This happens because the 561 dimensional space, even after PCA, has changing density, which is hard for DBSCAN.

None of the three methods split the data cleanly into six clusters matching the six activities. The main reason is that some activities, like sitting and standing, look very similar in sensor readings. Walking, walking upstairs, and walking downstairs also overlap a lot.

A strength of this experiment is that all three methods agree on the same broad two group split. A limitation is that finer activity differences are not captured well by any method without using the labels.

12 Conclusion

K-Means and Hierarchical clustering with Ward linkage gave the most meaningful and matching clusters, splitting the data into still and moving activities with a silhouette score close to 0.41. DBSCAN was more sensitive to the choice of ϵ and minPts and produced smaller, less label matching clusters, along with some noise points. Internal metrics best matched the visual PCA and t-SNE plots, since both agreed on two clear groups.

13 References

- [1] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, “A Public Domain Dataset for Human Activity Recognition Using Smartphones,” in Proc. 21st Eur. Symp. Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN), Bruges, Belgium, 2013.
- [2] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [3] UCI Machine Learning Repository, “Human Activity Recognition Using Smartphones Data Set,” archive.ics.uci.edu.